

Deterministic Optimization

Algebraic Aspects of Linear Optimization

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Basic Feasible Solution

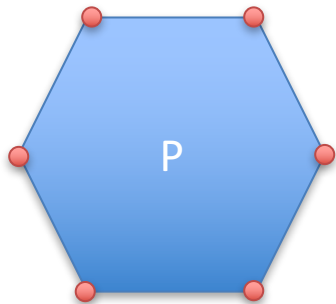
Algebraic Aspects of Linear Optimization

Learning Objective for this lesson

- Use algebraic language for linear optimization
 - Recognize algebraic description of extreme points

How to Compute an Extreme Point?

- Extreme points are corner points of a polyhedron.
- How can we find such points algebraically in a complicated polyhedron?



Easy to find by inspection

- $\sum_{j \in O(i)} f_{ij} - \sum_{j \in I(i)} f_{ij} = p_i \quad \forall i \in G$
- $\sum_{j \in O(i)} f_{ij} - \sum_{j \in I(i)} f_{ij} = -d_i \quad \forall i \in D$
- $\sum_{j \in O(i)} f_{ij} - \sum_{j \in I(i)} f_{ij} = 0 \quad \forall i \in N \setminus (G \cup D)$
- $f_{ij} = B_{ij}(\theta_i - \theta_j) \quad \forall (i, j) \in E$
- $-F_{ij} \leq f_{ij} \leq F_{ij} \quad \forall (i, j) \in E$
- $p_i^m \leq p_i \leq p_i^M \quad \forall i \in G$

Difficult to find by inspection

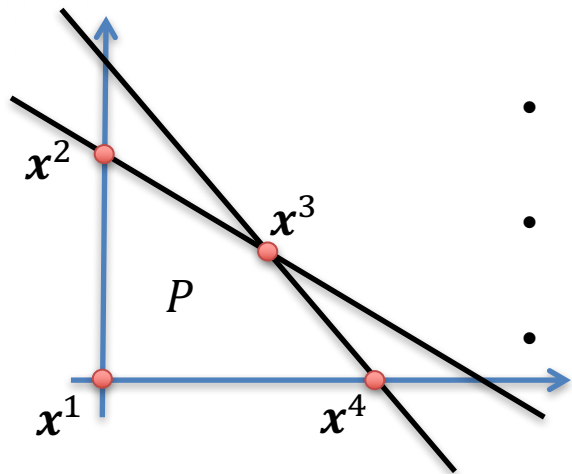
- Need an **algebraic** description and procedure to find extreme points.

Example

- Consider the following simple example:
 - $P = \{(x_1, x_2) \mid x_1 + x_2 \leq 3, x_1 + 2x_2 \leq 4, x_1 \geq 0, x_2 \geq 0\}$

Extreme points:

- x^1 is the solution of $x_1 = 0, x_2 = 0$
 - $x_1 = 0, x_2 = 0$ are **linearly indep active constraints** at x^1
- x^2 is the solution of $x_1 = 0, x_1 + 2x_2 = 4$
 - $x_1 = 0, x_1 + 2x_2 = 4$ are **l.i. active constraints** at x^2
- x^3 is the solution of $x_1 + x_2 = 3, x_1 + 2x_2 = 4$
 - $x_1 + x_2 = 3, x_1 + 2x_2 = 4$ are **l.i. active constraints** at x^3
- x^4 is the intersection of $x_1 + x_2 = 3, x_2 = 0$
 - $x_1 + x_2 = 3, x_2 = 0$ are **l.i. active constraints** at x^4



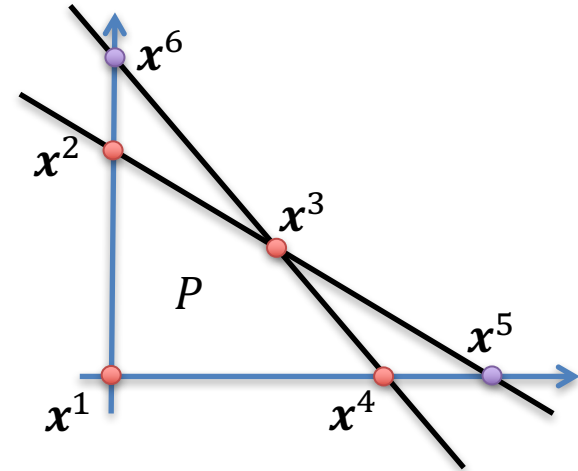
An extreme point is the **unique** solution of **n** linearly independent active constraints.

Key Concepts

- [Active constraints](#): A linear constraint that is satisfied as equality at a given point is said to be **active** or **binding** at that point. Otherwise, if an inequality constraint is satisfied as strict inequality at a point, it is called **inactive**.
- [Linear independent constraints](#): If the normal directions of two or more linear constraints are **linearly independent**, then these constraints are called linearly independent.
- [Linearly independent active constraints](#): Active constraints that are linearly independent.
- [Basic solution](#): The unique solution of n linearly independent active constraints in R^n .

Basic Feasible Solution = Extreme Points

- Basic feasible solution: A basic solution that is feasible.
 - $x^1, \dots, x^4$ are basic feasible solutions.
 - x^5, x^6 are basic solutions but not feasible.
- Basic feasible solution = extreme point
 - Every BFS is an extreme point.
 - Every extreme point is a BFS.
- Basic feasible solution is an algebraic description of extreme point.



Summary

- We learned:
 - An algebraic description of extreme point, namely basic feasible solution.
 - The solution of n linearly independent active constraints.
 - This provides a computational approach to find extreme points.